

Lorenzo Liuzzo

PHYSICIST · SOFTWARE ENGINEER · AI SYSTEMS

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Physicist and software engineer specialized in AI. Physics taught me to derive the model before writing the code; lately most of my time goes into learning how much of that writing can be delegated to a model, and where a person still has to decide.

Experience

MyThingsLab

🌐 MyThingsLab

FOUNDER, ARCHITECT AND DEVELOPER

July 2026 - Present

- A personal lab for AI-assisted development, started in July 2026: **more than 50 composable Python tools** in the first month, all on one SDK — *my-things-core* — exposing five contracts (ledger, policy, engine, GitHub, isolation) that every tool imports and no tool reimplements.
- What I am testing is control, not autonomy: headless workers pick an issue, do the deterministic pre-work with no model call, invoke the LLM once for the step needing judgement, and close it as a pull request — never a merge, so I keep the last word.
- The SDK is dependency-free and calls no model itself: it shells out to *gh* and *git*, and the LLM lives behind one *Engine* protocol whose deterministic default lets the whole fleet be exercised and tested without spending a token.

Money Ball AI

🌐 MoneyBallingAI

FOUNDER, TECHNICAL LEAD AND DEVELOPER

September 2025 - Present

- Founded and lead a team of four building one model that predicts the **exact box score** of an NBA game — every player and team line — with calibrated uncertainty. I own the architecture, from raw play-by-play to a calibrated prediction:
- **Data** — play-by-play parsed into periods, lineup stints and possessions, aggregated into cumulative stat windows and modelled as heterogeneous stint graphs with stable global identities for players and teams across seasons.
- **Encoder** — a heterogeneous GNN over those graphs with per-entity cross-game and hierarchical season/game/stint memory, pretrained on a box-score delta task for stable player and team strength embeddings.
- **Simulator** — a vectorized Monte Carlo possession engine whose policies are behavior-cloned on real possessions; **calibration** scores per-stat distributions against the actual box score with MAE, CRPS, PIT and coverage, under CI with a 134-test *pytest* suite.

Private Tutoring in Mathematics and Physics

Milan, Italy

IN-HOME TUTORING

2020 - Present

Continuous support for 20+ students: exam preparation, remedial tutoring and periodic reports to families.

Barista

Milan, Italy

DON SALVATORE - PIZZAIUOLO E OSTE

September - October 2024

Waiter

Milan, Italy

BAR TERRAZZA CLÉR

May - July 2022

Scout Leader

Milan, Italy

REPARTO BREITHORN - GRUPPO MI8 - AGESCI

2021 - 2022

Educational activities for youth aged 12-16: camps, outings, coordination with educators and families.

Education

University of Milan, Milan-Bicocca and Pavia

Milan/Pavia, Italy

MASTER'S DEGREE IN ARTIFICIAL INTELLIGENCE FOR SCIENCE AND TECHNOLOGY

September 2025 - Present

University of Milan

Milan, Italy

BACHELOR'S DEGREE IN PHYSICS, GRADE: 100/110

September 2021 - July 2025

Thesis: *ray tracing and Monte Carlo methods for uncertainty quantification in radio-telescope pointing for CMB observations.*

I.I.S. Severi-Correnti

Milan, Italy

SCIENTIFIC DIPLOMA, GRADE: 100/100

September 2016 - June 2021

EF International Language School

Brighton, England

FULL-IMMERSION INTENSIVE ENGLISH LANGUAGE COURSE

June - September 2019

Schools & Workshops

- **School on Quantum Simulation** — Department of Physics, University of Milan (9 - 11 September 2026).
- **Qiskit Fall Fest** — IBM Quantum · Department of Physics, University of Milan (31 October - 7 November 2025).

Projects

Gauge-Equivariant Graph Neural Networks

 [lorenzoliuzzo/EGNN](#)

DESIGNER/DEVELOPER

July - August 2026

Lattice Standard Model field theory simulated with gauge-equivariant GNNs in *PyTorch Geometric*: $SU(3) \times SU(2) \times U(1)$ gauge groups, Wilson-Dirac fermions and Higgs spontaneous symmetry breaking.

MSc Coursework Projects

 [lorenzoliuzzo](#)

DESIGNER/DEVELOPER

May - July 2026

- **Supervised Learning on Food Images** — a convolutional neural network under 10M parameters classifying the 251 classes of the FoodX-251 dataset.
- **Transverse Field Ising Model** — quantum phase transitions explored on *PennyLane*.
- **Unsupervised Learning on Country Data** — clustering and dimensionality reduction on socio-economic country-level data.

Telescope Pointing Simulation

 [lorenzoliuzzo/thesis-L30](#)

DESIGNER/DEVELOPER

June - July 2025

Julia model of a telescope's reflective system, propagating mirror defects to the pointing direction via *ray tracing* and *Monte Carlo* simulation.

Systems & Numerical Programming

C++ · Nim

DESIGNER/DEVELOPER

2023 - 2024

- **Ray Tracing Engine** — pair project at UNIMI: a physical model turned into structured, tested and documented numerical code, while rapidly learning a new language (*Nim*).
- **Dimensional Analysis Library** — “CTDA: Compile Time Dimensional Analysis”, C++ numeric types carrying units of measurement in the type system.

How I Build Software

- **AI across the whole workflow, not just the coding step.** Composable tools cover design, implementation, testing and documentation — which is how one person can cover more than 50 of them in a month, with *pytest*, *coverage* and *ruff* on every push.
- **Deterministic first, models only where judgement is needed.** The LLM sits behind a single *Engine* seam, so a tool's entire skeleton runs and is tested at zero token cost.
- **Cut complexity before it accrues.** The shared core stays dependency-free — shelling out to *gh* and *git* rather than pulling SDKs — and builds no abstraction for a need that does not exist yet.
- **Every side effect passes a gate.** A policy engine rules each action allow/ask/deny, tools open pull requests rather than merging, and every *ask* reaches me over Telegram before anything happens.
- **Derive the model, then write the code.** Physics first, then typed and documented numerical code — as in the telescope pointing thesis.

Skills

Languages

Italian, English C1

Programming Languages

C++, Python, Julia, C, Nim

AI Engineering

Claude, Claude Code, Agentic workflows, Prompt & context engineering

ML & Data Science

PyTorch, PyTorch Geometric, Scikit-learn, Pandas, NumPy, SciPy

Databases & Quantum

Neo4j, Cypher, PennyLane, Qiskit

Development Tools

Git, GitHub, GitHub Actions, pytest, ruff, Jupyter, LaTeX, Typst, Markdown